Supporting Information: Techno-economic Analysis Model

S1 Nomenclature

Indices are denoted as lower-case italic roman letters. Subscripts and superscripts are denoted as lower-case italic roman letters and upper-case italic roman letters, respectively.

Sets

$i \in \mathbf{I}$	Components
$k \in \mathbf{K}$	Reactions
$p \in \mathbf{P}$	Equipment
$s \in \mathbf{S}$	Stream

Subsets

$\mathbf{I}^{SA}$	Components of solvent and amine
$\mathbf{I}^{SC}$	System components
$\mathbf{I}^{NG}$	Components of solvent, amine, and salt ions
$\mathbf{I}^{NI}$	Components of solvent, amine, CO, and H_2
$\mathbf{I}^{PR}$	Components of CO and H_2

Parameters

A^{AN}/A^{CA}	Anode/cathode surface area
d^{ch}	Channel depth
$C_i^{CA,IN}$	Inlet component concentration at cathode side
C^{NEW}	Make-up cost for amine and electrolyte
$C^{T,SOL}$	Total cost for solvent and amine
$C^{T,MAT}$	Total cost for material

$C^{AMN,IN}$	Amine concentration before CO_2 capture
E^{REC}	Amount of required electricity for recycling stream
$\alpha^{AMN}/C^{T,AMN}$	Cost coefficient and total cost for amine
$\alpha^{AN}/C^{T,AN}$	Cost coefficient and total cost for anode
$\alpha^{ANL}/C^{T,ANL}$	Cost coefficient and total cost for anolyte
$\alpha^{CA}/C^{T,CA}$	Cost coefficient and total cost for cathode
α_i^{TREAT}	Treatment cost coefficient for components
$C^{T,CAR}$	Total cost for amine loss due to carbonate degradation
$\alpha^{CTL}/C^{T,CTL}$	Cost coefficient and total cost for catholyte
α^{ELEC}/C^{ELEC}	Cost coefficient and cost for electricity
$\alpha^{PL}/C^{T,PL}$	Cost coefficient and total cost for plastic
$\alpha^{MEM}/C^{T,MEM}$	Cost coefficient and total cost for membrane
$\alpha^{CASE}/C^{T,CASE}$	Cost coefficient and total cost for casing
D^{AN}/D^{CA}	Anode/ cathode depth
D^{CASE}	Depth of casing material
D^{PL}	Depth of plastic frame
ρ^{AN}	Anode density
ρ^{CA}	Cathode density
ρ^{PL}	Plastic density
z_k	Number of electrons per reaction
H	Height of electrochemical cell
K_i	Vapor-liquid phase equilibrium constants
L	Length of electrochemical cell
L^{AMN}	Amine loss from carbonate loss

T^{AMN}	Total amount of amine loss due to carbonate degradation
L^{CAR}	Carbonate loss
L^{AMN}	Amount of degraded amine per gram of captured CO_2
LF_p	Lifetime for each equipment
MA^{AMN}	Amine mass
$MA^{T,AMN}$	Total amine mass
MA^{AN}/MA^{CA}	Anode/ cathode mass
MA^{ANL}	Anolyte mass
MA^{CAL}	Catholyte mass
MW^{ANL}	Anolyte molecular weight
MW^{CAL}	Catholyte molecular weight
$MA^{T,ANL}$	Total anolyte mass
$MA^{T,CAL}$	Total catholyte mass
MA^{PL}	Plastic mass
N^{STACK}	Number of stack
N^{T}	Number of cell
E_k^{EQ}	Standard reversible reduction potential
$\Delta E_{k,k'}^{EQ}$	Difference of standard reversible reduction potential
ΔE	Potential difference between cathode and anode sides
$VE_{k,k'}$	Voltage efficiency
$EE_{k,k'}$	Energy efficiency
F	Faraday constant
F^{INST}/F^{MAINT}	Installation and maintenance cost factor

FE_k^{CA}	Faradaic efficiency for reaction in cathode side
$\nu_{k,i}$	Stoichiometric coefficient of component *i* in reaction *k*
P^{ATM}	Atmospheric pressure
Q^{AMN}	Total amount of amine required
Q^{CAP}	Total amount of CO_2 captured
Q^{UP}	CO_2 uptake
T^{RM}	Room temperature
χ^{AMN}	Amine conversion
v^{ELY}	Velocity of electrolyte
$V^{CA,FLOW}$	Volumetric flowrate in cathode side
V^{ANL}/V^{CAL}	Volume of anolyte/catholyte
V^{AN}/V^{CA}	Volume of compartments in anode/cathode side
V_i^{FEED}	Component flowrate for vapor feed
L	Liquid flowrate
V	Vapor flowrate
P^{CELL}	Electric power for a single cell
W^{CELL}	Electric energy for a single cell
X_i	Component molar fraction in liquid phase
Y_i	Component molar fraction in vapor phase
Z_i^{FEED}	Component molar fraction in the liquid phase of electrochemical cell outlet stream
Y_i^{FEED}	Component molar fraction in the vapor phase of electrochemical cell outlet stream
η^{AMN}	Amine degradation fraction

Non-negative Continuous Variables

$A^{T,MEM}$	Total membrane area
$C_i^{CA,OUT}$	Outlet component concentration at cathode side
C^{CAP}/C^{OP}	Capital and operating cost
C^{MAINT}/C^{NEW}	Maintenance and make-up cost
$C^{RECYCLE}$	Recycle cost
C^{TREAT}	Treatment cost
J_k^{CA}	Current density for reaction *k* in cathode side
J^T	Total current density
E^{CA}	Potential at cathode side
E^{RE}	Reference potential
$E^{CA,RE}$	Relative potential between cathode and reference electrode
E^{AN}	Potential at cathode side
E^{RE}	Reference potential
$E^{AN,RE}$	Relative potential between cathode and reference electrode
η_k^{AN}	Overpotential at anode side
η_k^{CA}	Overpotential at cathode side
$C_i^{CA,IN}$	Inlet component concentration at the cathode side
$C_i^{CA,OUT}$	Outlet component concentration at the cathode side
I	Electric current
MA^{CASE}	Total case mass
$MA^{T,AN}$	Total anode mass
$MA^{T,CA}$	Total cathode mass
$MA^{T,PL}$	Total plastic mass

ξ_k^{CA}	Reaction rate at the cathode side
$F^{MOL,P}$	Molar flowrate of product
$F_{s,i}^{MOL}$	Molar flowrate of component *I* in stream *s*
$F^{MA,P}$	Mass flowrate of product
$F^{MA,T}$	Total mass flowrate of product
W^T	Total electric energy
X_i^{CA}	Molar composition at the cathode side

Symbols

COER	CO evolution reaction
CARB	Carbamate
HER	Hydrogen evolution reaction
OER	Oxygen evolution reaction
SAS	Solvent, amine, and salt

S2 Electrochemical System Model

The techno-economic analysis is based on a system model for integrated carbon capture and electrochemical conversion along with cost calculations. The modeled system is described within the grey box of Figure S1.

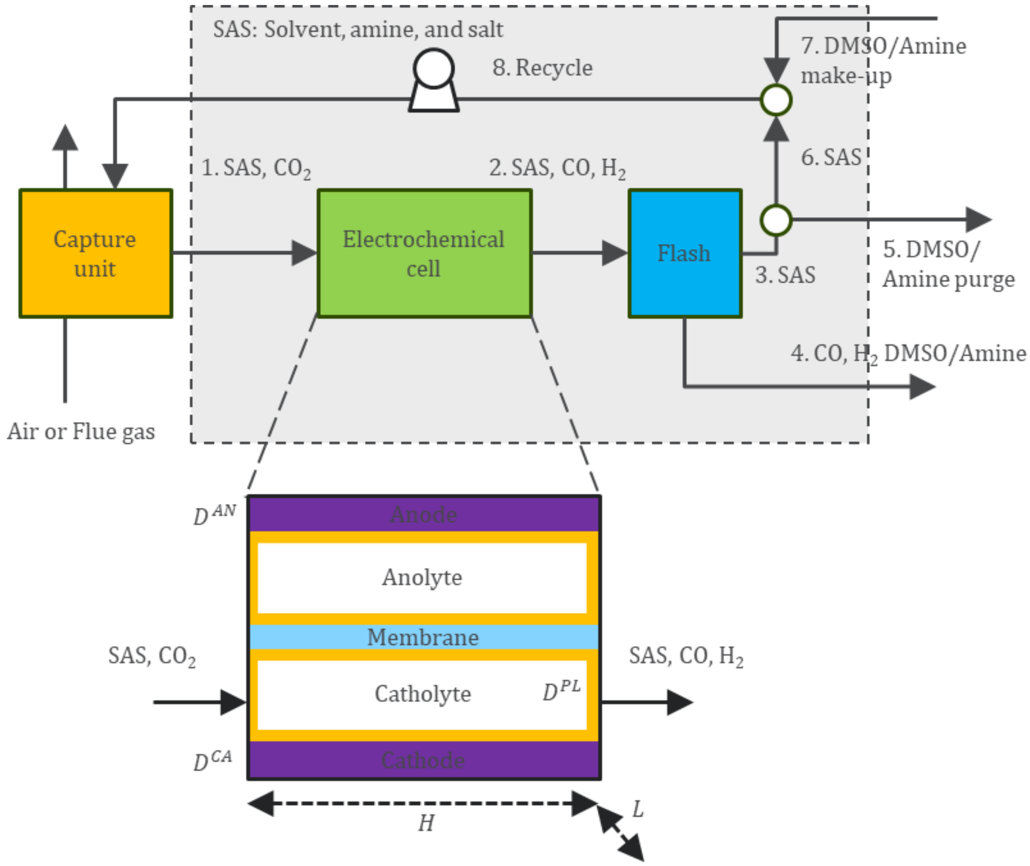

Figure S1. System description for the integrated CO_2 capture and electrochemical reduction. The modeled system is described within the grey box. The process stream with captured CO_2 enters the system boundary and the regenerated amine/solvent stream is recycled to the CO_2 capture unit.

As shown in Figure S1, the outlet stream (stream 1 comprising solvent, amine, salt, and carbamic acid) from capture unit enters the electrochemical cell. After electrochemical reduction of CO_2, the outlet stream (stream 2 comprising solvent, amine, salt, carbamic acid, CO, and H_2) enters a flash unit to separate the liquid-phase stream (stream 3 comprising solvent, amine, carbamic acid, and salt) from the vapor-phase stream (stream 4 comprising CO, H_2). Some amount of solvent and amine can be entrained into the vapor-phase, which results in solvent and amine loss. Certain amount of amine is degraded, and purging (stream 5) is used to address degradation. The remaining liquid-phase stream (stream 6) is recycled into the capture unit with additional amine and solvent (stream 7) to make-up for their loss from the system. We use ASPEN simulation to obtain the vapor-liquid equilibrium constant at the flash separator. The details of this procedure are given in Section S3.3 of Supporting Information.

S3 Techno-economic Model

We develop a bottom-up techno-economic model for integrated carbon capture and electrochemical conversion, starting from component material costs and building up to system-level cost metrics, following the DOE/NETL guidelines and our previous TEA studies [1-4]. The model accounts for electrochemical cell design, material requirements, electricity consumption, solvent and amine losses, and vapor–liquid equilibrium in the flash separation unit. Electrochemical cells are arranged in stacks, modules, and trains to achieve the target system size, with cost scaling handled through numbering-up and standard cost-scaling exponents. Capital cost estimation proceeds from the installed costs to obtain the Engineering, Procurement, and Construction Cost (EPCC) by applying contractor costs. Application of process and project contingencies yields the Total Plant Cost (TPC). Conversion to Total Overnight Cost (TOC) follows multipliers specified in NETL cost estimation guidelines. Operating costs were separated into fixed and variable components, with the latter including electricity consumption, solvent and amine makeup, recycling, and waste treatment. Together, these contributions are used to estimate the levelized cost of product (LCOP), reported as the sum of the levelized total ownership cost (LTOC) and levelized fixed and variable operating and maintenance cost (LTFOM and LTVOM). See section S3.5 for detailed descriptions and definitions.

The technoeconomic model is built on an underlying process model, which couples a physics-based, zero-dimensional representation of the electrochemical cell, a vapor–liquid equilibrium model of the flash unit, and a mass and energy balance model of all balance of plant components. The process model provides product flowrates, current densities, and solvent/amine losses that serve as inputs to the TEA, while the TEA translates these physical outputs into capital, operating, and levelized production costs. Full cost equations, parameter values, and intermediate outputs are provided in S3.5 and the subsequent tables.

Modeling implementation and key assumptions are as follows: (1) we size the unit based on our process model, and use a combination of literature, experimental and Aspen simulation data to estimate phase equilibrium in the flash separator, and degradation rates; (2) we neglect the effect of contaminants on electrochemical performance; (3) our make-up volume is based on amine/solvent degradation rate as well as amine/solvent entrained vapor loss; (4) the solvent loss by vapor entrainment is estimated from vapor-liquid equilibrium calculations implemented in our model; (5) we further consider that electrochemical cells are arranged in stacks, and stacks into modules, and modules combined (additively) into trains up to target system size; steel plates encase the stacks; (6) multiple stacks constitute a module, and multiple modules constitute the full electrochemical train; (7) each electrochemical cell is identical in design, feed, and operating conditions.

S3.1 Parameter Calculation

For a parallel plate cell assumed in this analysis, all components have the same cross-sectional area (A). The cell comprises cathode (CA), membrane (MEM), anode (AN), and plastic (PL) spacers for the anolyte and catholyte channels. The channel is a hollow enclosure of plastic of width d^{ch} and void fraction ϵ^{ch}. The list of design parameters for the electrochemical cell is given in **Table S1**. With these design parameters, we further calculate other parameters. The mass for plastic (MA^{PL}) is determined using equations S1 to S3:

$$MA^{PL} = AD^{PL}\rho^{PL} \tag{S1}$$

Equations S4 to S6 respectively describe the calculation of channel volume, cuboidal channel cross-sectional area with height H and length L, and volumetric flow rate.

$$V^{AN} = V^{CA} = HLd^{ch} \tag{S2}$$

$$A = A^{AN} = A^{CA} = HL \tag{S3}$$

$$V^{CA,FLOW} = Hv^{ELY}d^{ch} \tag{S4}$$

Table S1. List of design parameters for electrochemical cell from literature [5]

Parameters	Definition	Values
H	Height of electrochemical cell (cm)	160

L	Length of electrochemical cell (cm)	160
D^{PL}	Depth of plastic frame (cm)	0.6
D^{CASE}	Depth of casing material (cm)	0.5
d^{ch}	Channel depth (cm)	0.5
V^{ANL}/V^{CAL}	Volume of anolyte/catholyte (cm^3)	15
v^{ELY}	Velocity of electrolyte (cm/s)	6.5

Electric Power Calculation: Electric current is determined using the product of total current density and cross-sectional area (Equation S12). Electric power is determined as the product of potential difference and electric work (Equation S13). Here, J^T, E^{AN}, E^{CA} are parameters obtained from experiment, and represent total current density, anode potential and cathode potential, respectively. The complete list of parameters obtained from the experiment is given in **Table S2**.:

$$I = J^T A \tag{S5}$$

$$P^{CELL} = (E^{AN} - E^{CA}) \times I \tag{S6}$$

Table S2. List of input parameters from experiment

Parameters	Definition
FE_k^{CA}	Faradaic efficiency for reaction in cathode side
$E^{AN,RE}$	Relative potential between anode and Ag (V)
$E^{CA,RE}$	Relative potential between cathode and Ag (V)
J^T	Total current density
E^{RE}	Reference potential (0.72 V)
Z_i^{FEED}	Component molar fraction in the liquid phase of electrochemical cell outlet stream
Y_i^{FEED}	Component molar fraction in the vapor phase of electrochemical cell outlet stream

Q^{CAP}	Total amount of CO_2 captured (2637 mol/L)
Q^{UP}	CO_2 uptake
L^{CAR}	Carbonate loss
χ^{AMN}	Amine conversion (0.9)
$C^{AMN,IN}$	Amine concentration before CO_2 capture (0.3 mol/L)

Electrolyte and Amine Mass: We first determine the mass of electrolytes (MA^{ANL} and MA^{CAL}) for the case when cell is filled (i.e. one residence time worth of flowrate) and the mass of dissolved species present in those electrolytes as shown in equations S7 to S8. The complete list of input parameters from literature is given in **Table S3**:

$$MA^{ANL} = \rho^{ANL} V^{ANL} \tag{S7}$$

$$MA^{CAL} = \rho^{CAL} V^{CAL} \tag{S8}$$

$$MA^{AMN} = MW^{AMN} C^{AMN,IN} V^{CA} \tag{S9}$$

Table S3. List of input parameters from literature [6]

Parameters	Definition	Values
E_{COER}^{EQ}	Standard reversible reduction potential of COER (V)	-0.22
E_{HER}^{EQ}	Standard reversible reduction potential of HER (V)	0
E_{OER}^{EQ}	Standard reversible reduction potential of OER (V)	1.23
ρ^{AN}	Anode density	NIST
ρ^{CA}	Cathode density	NIST
ρ^{ANL}	Anolyte density	NIST
ρ^{CAL}	Catholyte density	NIST

ρ^{CASE}	Steel density (g/cm^3)	7.96
MW^{AMN}	Amine molecular weight	NIST
MW^{ANL}	Anolyte molecule weight	NIST
MW^{CAL}	Catholyte molecule weight	NIST

S3.2 Electrochemical Cell

Partial Current Density and Potentials: Based on faradaic efficiency at cathode side FE_k^{CA}, which describes the amount of current supplied for reaction *k*, and total current density J^T, we use equation S10 to determine the partial current density. We equations S11 and S12 to determine the potential at anode and cathode sides, respectively:

$$J_k^{CA} = FE_k^{CA} \times J^T \tag{S10}$$

$$E^{CA} = E^{CA,RE} + E^{RE} \tag{S11}$$

$$E^{AN} = E^{AN,RE} + E^{RE} \tag{S12}$$

Overpotentials: We use equations S13 and S14 to determine the overpotential at anode and cathode sides, respectively:

$$\eta_k^{CA} = E_k^{EQ} - E^{CA} \tag{S13}$$

$$\eta_k^{AN} = E^{AN} - E_k^{EQ} \tag{S14}$$

Potential Difference: We use equation S15 to determine the equilibrium potential (standard reversible reduction potential) difference ($\Delta E_{k,k'}^{EQ}$) and use equation S16 to determine the potential difference between anode and cathode sides (ΔE):

$$\Delta E_{k,k'}^{EQ} = E_k^{EQ} - E_{k'}^{EQ} \tag{S15}$$

$$\Delta E = E^{AN} - E^{CA} \tag{S16}$$

Voltage Efficiency and Energy Efficiency: We use equation S17 to determine the voltage efficiency ($VE_{k,k'}$) and use equation S18 to determine energy efficiency ($EE_{k,k'}$):

$$VE_{k,k'} = \frac{\Delta E^{EQ}_{k,k'}}{\Delta E} \tag{S17}$$

$$EE_{k,k'} = FE_k VE_{k,k'} \tag{S18}$$

Reaction Rate and Outlet Component Concentration: We use equations S19 and S20 to determine reaction rate (ξ_k^{CA}) for each electrochemical cell and outlet molar flowrate ($F_{2,i}^{MOL}$), respectively [7-8]:

$$\xi_k^{CA} = \frac{FE_k^{CA} I}{z_k F} \tag{S19}$$

$$F_{2,i}^{MOL} = F_{1,i}^{MOL} + N^T \sum_k \nu_{k,i} \xi_k^{CA} \tag{S20}$$

Electrolyte Molar Flowrate: Molar flowrate for electrolyte in the stream 2 ($F_{2,\mathrm{CAL}}^{MOL}$) is determined based on amine concentration:

$$F_{2,\mathrm{CAL}}^{MOL} = \frac{\rho^{CAL} F_{2,\mathrm{AMN}}^{MOL}}{C^{AMN,IN} MA^{\mathrm{CAL}}} \tag{S21}$$

Product Flowrate: We use equations S22, S23, and S24 to determine product molar flowrate ($F^{MOL,P}$), product mass flowrate ($F^{MA,P}$), and total product mass flowrate ($F^{MA,T}$ with the unit of ton per year), respectively:

$$F^{MOL,P} = F_{2,\mathrm{CO}}^{MOL} \tag{S22}$$

$$F^{MA,P} = F_{2,\mathrm{CO}}^{MOL} MW_{CO} \tag{S23}$$

$$F^{MA,T} = 365 \times 24 \times 3600 \times 10^{-6} F^{MA,P} \tag{S24}$$

Here, we fix $F^{MA,P}$ as 1 kg/s

CO_2/ Carbamate Conversion: CO_2 conversion serves as an indicator to evaluate the performance of electrochemical cell. Since the stoichiometric ratio between CO_2 and carbamate is one, CO_2 conversion is equal to **carbamate** conversion. To determine CO_2 conversion, we firstly determine **carbamate** conversion using equation S25:

$$X_{CO_2}^{CA} = X_{\mathrm{CARB}}^{CA} = 1 - \frac{F_{2,\mathrm{CARB}}^{MOL}}{F_{1,\mathrm{CARB}}^{MOL}} \tag{S25}$$

S3.3 Flash Model

We use a flash model to represent and model vapor-liquid phase equilibrium, and consequently, solvent and amine loss due to vaporization within the system.

Component Molar Balance: The molar feed flowrate of flash operation consists of molar flowrate for catholyte and amine (in liquid phase) and molar flowrate for CO and H_2 (in vapor phase) at the outlet stream (stream 2) of electrochemical cell. Following vapor-liquid phase equilibrium, the overall mixed stream will be split into a liquid-phase stream ($F_{3,i}^{MOL}$) and a vapor-phase stream ($F_{4,i}^{MOL}$).

$$F_{2,i}^{MOL} = F_{3,i}^{MOL} + F_{4,i}^{MOL} \tag{S26}$$

Calculation of Molar Composition: The component molar compositions in the liquid-phase stream (stream 3) and vapor-phase stream (stream 4) are determined using equations S26 and S27, respectively.

$$X_i \sum_{i \in \mathbf{I}^{NG}} F_{3,i}^{MOL} = F_{3,i}^{MOL} \tag{S27}$$

$$Y_i \sum_{i \in \mathbf{I}^{NI}} F_{4,i}^{MOL} = F_{4,i}^{MOL} \tag{S28}$$

Equilibrium Relationship: We consider the equilibrium constants K_i as parameters from ASPEN flash simulation, where the input feed composition Z_i^{FEED} is directly obtained from experiment measurement. Hence, the relationship of vapor-phase composition and liquid-phase composition is a linear relationship. In ASPEN simulation, we used electrolyte NRTL property method (ENRTL-RK) as the property method to determine K_i, which is the output of ASPEN simulation.

$$Y_i = K_i X_i, \quad i \in I^{SA} \tag{S29}$$

Moreover, the summation of component mole fraction is equal to one in both vapor and liquid phases:

$$\sum_{i \in \mathbf{I}^{NG}} X_i = 1, \quad X_{CO} = 0, \quad X_{H_2} = 0 \tag{S30}$$

$$\sum_{i \in \mathbf{I}^{NI}} Y_i = 1 \tag{S31}$$

S3.4 Other Model Constraints

In this Section, we describe the remaining model constraints, including determination of number of stacks, total amount of electricity consumption, total cell component mass, total electrolyte and amine mass, as well as molar balances around the mixer and splitter.

Number of Stacks: Equation S34 estimates the number of stacks:

$$N^{STACK} = \frac{500}{N^T} \quad \text{(S32)}$$

We consider the number of cells per stack as 500 [5]. If the number of cells is less than 500, then the stack number is equal to 1.

Total Amount of Electricity: Total electricity consumption is then calculated as the product of the total number of cells, and the power consumption per cell, as shown in equation S14:

$$W^T = N^T P^{CELL} \quad \text{(S33)}$$

Total Cell Component Mass: We use equations S36 – S37 to determine the total mass for plastic, and membrane material:

$$MA^{T,PL} = N^T MA^{PL} \quad \text{(S34)}$$

$$A^{T,MEM} = N^T A \quad \text{(S35)}$$

For each stack, since there are two (stainless steel) compression plates supporting the ends [5], we determine the case mass as described in equation S38:

$$MA^{CASE} = N^{STACK}(2D^{CASE} A\rho^{CASE}) \quad \text{(S36)}$$

Here, ρ^{CASE} is the steel density, which is obtained from NIST.

Total Electrolyte and Amine Mass: We estimate the total mass for anolyte, catholyte, and amine as shown in equations S23 to S25:

$$MA^{T,ANL} = N^T MA^{ANL} \quad \text{(S37)}$$

$$MA^{T,CAL} = N^T MA^{CAL} \quad \text{(S38)}$$

$$MA^{T,AMN} = N^T MA^{AMN} \quad \text{(S39)}$$

Solvent and Electrolyte Degradation: We use and model a purge stream (stream 5) to determine solvent and electrolyte loss from the liquid-phase outlet stream from flash (stream 2) due to degradation:

$$F_{2,i}^{MOL} = F_{5,i}^{MOL} + F_{6,i}^{MOL} \quad \text{(S40)}$$

The amount of degraded amine ($F_{5,\mathrm{AMN}}^{MOL}$) is related to the amount of captured CO_2:

$$F_{5,\mathrm{AMN}}^{MOL} = L^{\mathrm{AMN}} \frac{(F_{1,\mathrm{CARB}}^{MOL} - F_{2,\mathrm{CARB}}^{MOL}) MA^{\mathrm{CO_2}}}{MA^{\mathrm{AMN}}} \tag{S41}$$

Molar flowrate for electrolyte in the purge stream ($F_{5,\mathrm{CAL}}^{MOL}$) is determined based on amine concentration:

$$F_{5,\mathrm{CAL}}^{MOL} = \frac{\rho^{CAL} F_{5,\mathrm{AMN}}^{MOL}}{C^{AMN,IN} MA^{\mathrm{CAL}}} \tag{S42}$$

Molar flowrate for carbamic acid in the purge stream ($F_{5,\mathrm{CARB}}^{MOL}$) is determined based on amine molar flowrate and amine conversion:

$$F_{5,\mathrm{CARB}}^{MOL} = \frac{\chi^{AMN}}{1-\chi^{AMN}} F_{5,\mathrm{AMN}}^{MOL} \tag{S43}$$

Here, L^{AMN} is the amount of degraded amine (gram) per gram of captured CO_2, which is 1.4×10^{-3} g per captured CO_2 [20]. χ^{AMN} is the amine conversion.

Molar Balance at the Capture Unit: We model the molar balance of capture unit for carbamic acid and amine using equations S46 and S47, respectively:

$$F_{1,\mathrm{CARB}}^{MOL} = F_{8,\mathrm{CARB}}^{MOL} + \chi^{AMN} F_{8,\mathrm{AMN}}^{MOL} \tag{S44}$$

$$F_{1,\mathrm{AMN}}^{MOL} = F_{8,\mathrm{CARB}}^{MOL} - \chi^{AMN} F_{8,\mathrm{AMN}}^{MOL} \tag{S45}$$

Molar Balance at the Mixer: The molar balance for the mixer connecting inlet stream 6 and make-up stream (stream 7) and outlet stream 8 is determined using equation S48.

$$F_{8,\mathrm{i}}^{MOL} = F_{6,i}^{MOL} + F_{7,i}^{MOL} \tag{S46}$$

Molar flowrate $F_{7,i}^{MOL}$ is determined as follows:

$$F_{7,i}^{MOL} = F_{4,i}^{MOL} + F_{5,i}^{MOL}, \ i \in \mathbf{I}^{SA} \tag{S47}$$

$$F_{7,\mathrm{CARB}}^{MOL} = 0 \tag{S48}$$

S3.5 Cost calculations

Total Material Cost: We model solvent as an initial, one-time purchase plus make-up stream to offset the degradation that happens over time as it circulates within the system. Material costs consist of costs of cathode, anode, plastics, membrane, and casing materials [5].

Table S4. List of material price

Cost parameters	Price
α^{AN} ($ per m^2)	320.21 [9]
α^{Ag} ($ per m^2)	26351.36 [9]
α^{Zn} ($ per m^2)	576 [10]
α^{PL} ($ per kg)	15 [11]
α^{MEM} ($ per m^2)	2777.78 [12]
α^{CASE} ($ per kg)	1.5 [13]
$\alpha^{ANL}/\alpha^{CTL}$ ($ per kg)	2.34 [14]
α^{AMN} ($ per kg)	0.86 [15]

All system components cost estimates are on a mass-basis, except for membranes which use the cross-sectional area as basis. Equations S51 to S58 describe how we determine costs for anode, cathode, plastic, membrane, anolyte, catholyte, amine, and stack casing for the train of electrochemical cell. Stack casing is to support each stack. We obtain the cost coefficients of anode (α^{AN}) and cathode (α^{CA}) material from Goodfellow [9] and Arbor Scientific [10]. We assume that the plastic material used within the electrochemical cell channel is the common material, polytetrafluoroethylene (PTFE) [11], the membrane material used with the electrochemical cell is Nafion N117 [12], and the casing material is steel [13]. We use dimethyl sulfoxide (DMSO) as anolyte and catholyte with the DMSO price obtained from Alibaba [14]. The list of material prices is given in Table S4.

$$C^{T,AN} = MA^{T,AN}\alpha^{AN} \tag{S49}$$

$$C^{T,CA} = MA^{T,CA}\alpha^{CA} \tag{S50}$$

$$C^{T,PL} = MA^{T,PL}\alpha^{PL} \tag{S51}$$

$$C^{T,MEM} = A^{T,MEM}\alpha^{MEM} \tag{S52}$$

$$C^{T,ANL} = MA^{T,ANL}\alpha^{ANL} \tag{S53}$$

$$C^{T,CTL} = MA^{T,CTL}\alpha^{CTL} \tag{S54}$$

$$C^{T,AMN} = MA^{T,AMN}\alpha^{AMN} \tag{S55}$$

$$C^{T,CASE} = MA^{T,CASE}\alpha^{CASE} \tag{S56}$$

The total cost for solvent and amine is determined by summing anolyte, catholyte and amine costs as shown in equation S57:

$$C^{T,SOL} = C^{T,ANL} + C^{T,CTL} + C^{T,AMN} \tag{S57}$$

The total cost for material is determined from the summation in S58:

$$C^{T,MAT} = C^{T,CA} + C^{T,AN} + C^{T,PL} + C^{T,MEM} + C^{T,CASE} \tag{S58}$$

Annualized Capital Cost: We determine bare erected cost (BEC) from the annualized total capital equipment cost (C^{CAP}) using the factorial method based on Lang Factors [16]. F^{INST} is the installation cost factor and is taken as 0.1 [17]. Note that this approach should be used only in the earliest stages of process design and in the absence of detailed design information. For the reference case estimates, we approximate annualized capital cost based on the lifetime for each equipment p (L_p), which is 5 years for the cathode (Ag, Zn) and membrane, and 20 years for the anode (Pt, Pd), and other electrolyzer components [18]. Subsequently, we determine the Engineering Procurement & Construction Costs (EPCC), which describes the installed cost of equipment including contractor costs; the Total Plant Cost (TPC), which captures project and process contingencies, representing the total estimated cost to build the plant, and the Total Overnight Cost (TOC), which adjusts the TPC for owner's costs, preproduction costs, and financing assumptions, representing the full capital investment required if the plant were built "overnight." See Table

$$C^{CAP} = \sum_p \frac{C^{p,All}}{L_p} \tag{S59}$$

$$BEC = (1 + F^{INST})C^{CAP} \tag{S60}$$

$$EPCC = BEC(1 + f^{count})$$

$$TPC = EPCC\left(1 + f^{proc} + f^{proj}\right)$$

$$TOC = TPC\left(1 + f^{preprod} + f^{inventory} + f^{\frac{owner}{other}}\right)$$

Table S5: Capital Cost Estimation Parameters [12, 19-24]

Parameter	Units	Value
f_{TOC}^{TASC}		1.07
f^{FCF}		0.06
f^{proc}	% of EPCC	25[1]
f^{proj}	% of EPCC	20
f^{count}	% of BEC	9
$f^{preprod}$	% of TPC	2
$f^{\frac{owner}{other}}$	% of TPC	15
$f^{inventory}$	% of TPC	0.5
f^{CAP}		0.907185
Economic lifetime (N)	Years	20

Electricity Cost: We consider the base electricity price α^{ELEC} as \$0.06 per kWh [5, 12]. For the sensitivity analysis to investigate the influence of electricity price on electrochemical reduction cost, we considered α^{ELEC} between \$0.01 per kWh and \$0.06 per kWh. Equation S58 estimates the total energy cost:

$$C^{ELEC} = \alpha^{ELEC} W^T \tag{S61}$$

Maintenance Cost: We determine the annualized maintenance cost as comprising two components: the replacement cost for the membrane over the economic life of the plant, and the

[1] Based on AACE recommendations for new technologies with limited (bench-scale) data

overall maintenance cost for installed equipment, calculated as a product of their installed capital costs and a maintenance cost factor (F^{MAINT}) with a value of 0.025 [18] as shown in S59.

$$C^{MAINT} = \left(\frac{1}{N}\right)(C^{T,MEM})\left(\frac{N}{n} - 1\right) + F^{MAINT} C^{CAP} \quad \text{(S62)}$$

Make-Up Cost: The system requires make-up supplies due to amine degradation, and loss of amine and electrolyte due to vapor entrainment. The vapor loss is estimated using our custom VLE model.

$$C^{MKP} = \alpha^{AMN} F_{7,AMN}^{MOL} + \alpha^{CAL} F_{7,CAL}^{MOL} \quad \text{(S63)}$$

Recycle Cost: The system requires the recycling of liquid-phase stream as shown by stream 8. Stream recycling requires electricity investment for pumping. We determine the recycling cost based on the electricity cost required for recycling stream 8:

$$C^{RECYCLE} = \alpha^{ELEC} E^{REC} \sum_{i \in \mathbf{I}^{SC}} F_{8,i}^{MOL} \quad \text{(S64)}$$

Here, E^{REC}=2700 J per mol, which represents the amount of pumping power per mol of recycle stream and determined through an ASPEN simulation [25].

Waste Treatment Cost: The purge stream (stream 5) comprises effluent solvent and amine streams classified as toxic waste, which requires treatment cost. We determine treatment cost based on the flowrate of purge stream (stream 5) and α_i^{TREAT} represents treatment cost coefficient for component *I* [26,27]:

$$C^{TREAT} = \sum_{i \in \mathbf{I}} \alpha_i^{TREAT} F_{5,i}^{MOL} \quad \text{(S65)}$$

Variable Operating and Maintenance Cost (TVOM): The variable operating cost consists of electricity cost, recycling (pumping) cost, and make-up cost and waste treatment costs, as shown in equation S65 in \$/year.

$$TVOM = C^{ELEC} + C^{RECYCLE} + C^{MKP} + C^{TREAT} \quad \text{(S66)}$$

Fixed Operating and Maintenance Cost: (TFOM) The fixed operating cost consists contributions from labor, supervision, supplies, taxes, insurance and maintenance, in \$/year.

$$C^{LA} = f^{LA} C^{EQUIP} \quad \text{(S67)}$$

$$C^{SU} = f^{SU} C^{LA} \tag{S68}$$

$$C^{SP} = f^{SP} C^{MAINT} \tag{S69}$$

$$C^{TX} = f^{TX} EPCC \tag{S69}$$

$$C^{IU} = f^{IU} EPCC \tag{S70}$$

$$TFOM = C^{LA} + C^{SU} + C^{SP} + C^{TX} + C^{IU} + C^{MAINT} \tag{S71}$$

Levelized Cost of Product (LCOP): We express the levelized cost of product (LCOP) as the sum of the levelized total ownership cost (LTOC), levelized fixed operating and maintenance cost (LFO&M), and levelized variable operating and maintenance cost (LVO&M). The capital component, LTOC, is obtained by multiplying the total overnight cost (TOC) by two factors: the total-as-spent-to-TOC multiplier, which adjusts for escalation from overnight to as-spent dollars, and the fixed charge factor (FCF). The FCF annualizes the investment and represents the capital recovery requirement, accounting for the assumed rate of return, financing structure, plant life, and capacity factor. Together, these terms ensure that both capital and operating costs are normalized to the annual production of CO, providing a consistent metric for economic comparison

$$LTVOM = \frac{TVOM}{Annual\ Production\ Rate} \tag{S71}$$

$$LTFOM = \frac{TFOM}{Annual\ Production\ Rate} \tag{S71}$$

$$LTOC = \frac{\left(TOC * f_{TOC}^{TASC} * f^{FCF}\right)}{Annual\ Production\ Rate} \tag{S71}$$

$$LCOP = LTVOM + LTFOM + LTOC \tag{S71}$$

S4 Process Stream and Cost Breakdown for a Reference Case

Table S6: Process Stream Table

kg/s	1	2	3	4	5	6	7	8
H2O	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

R-NHCOOH	4.995	1.249	1.249	0.000	0.034	1.215	0.000	1.215
R-NH_2	0.244	2.422	2.422	0.000	0.002	2.419	0.002	2.422
DMSO	145.375	145.375	145.375	0.000	0.132	145.243	0.132	145.375
H2	0.000	0.011	0.000	0.011	0.000	0.000	0.000	0.000
CO	0.000	1.000	0.000	1.000	0.000	0.000	0.000	0.000

Table S7: Material & Technical Specifications

Cathode material	-	Zn
Anode material	-	Ni
Membrane material	-	Nafion N117[2]
Electrolyte	-	0.3M MEA in DMSO
Total Electrode Area	m2	58, 562
Current Density	mA/cm2	13.56
CO Faradaic Efficiency	%	0.84
Overall Cell Potential	V	3.61
Membrane lifetime	Years	7
# Cells per stack	#	500[3]

Table S8: Cost Breakdown Table

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CO_2 Inlet Flowrate	Mt/year	55.50
CO_2 Outlet Flowrate	Mt/year	5.55
CO Production Rate	Mt/year	31536
Hydrogen Production Rate	Mt/year	347.59
Anode	$	18,752,526.46
Cathode	$	33,731,896.65
Plastic	$	11,595,339.47
Membrane	$	162,627,564.24
Casing	$	13,984,682.15
Total Equipment Cost	$	240,692,008.97
CAPEX	$/Mt	1,221.17
Maintenance	$/Mt	213.70
Variable O&M Costs	$/Mt	850.41
Electricity	$/Mt	477.79
Solvents & Chemicals Make-up	$/Mt	353.09
Waste Treatment	$/Mt	19.44
Recycle	$/Mt	0.09
Total Cost	$/Mt	2,285.27
Total Cost (Revenue Adjusted)	$/Mt	2,007.92

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